

Industrial Internship Report on Food Delivery Application

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Executive Summary

My project was a Food Delivery Application developed using React, TypeScript, Tailwind CSS, and Supabase PostgreSQL. The application acts as a bridge between customers, restaurant owners, delivery partners, and administrators through a modern web platform.

The system allows customers to browse restaurants, search food items, add products to cart, place orders, track deliveries in real-time, manage favourites, and submit reviews. Restaurant owners can manage menus, accept and process orders, approve delivery partners, and monitor restaurant operations through a dedicated dashboard. Delivery partners can accept delivery requests, update delivery status, and complete deliveries, while administrators can monitor users, restaurants, and platform analytics.

The project was designed with role-based authentication, real-time notifications, responsive UI design, and secure database management using Supabase services. This internship project helped me gain practical exposure to full stack web application development, database integration, authentication systems, dashboard management, and real-world software engineering workflows.

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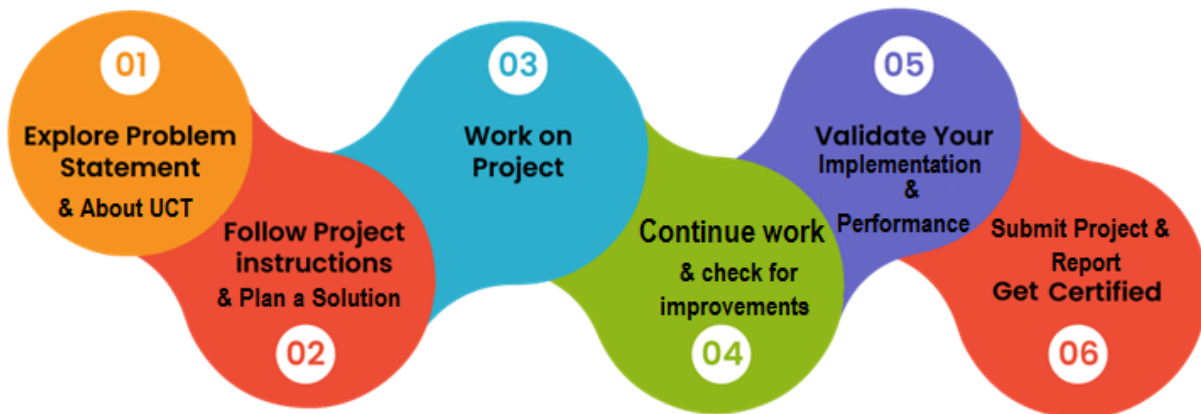
1 Preface

The Industrial Internship Program provided me with valuable practical exposure to real-world software development and problem-solving. During this internship, I worked on a Food Delivery Application project assigned by UpSkill Campus and UniConverge Technologies Pvt Ltd (UCT).

The project was developed using React, TypeScript, Tailwind CSS, and Supabase PostgreSQL. The application connects customers, restaurant owners, delivery partners, and administrators through a single platform for food ordering and delivery management.

Throughout the internship, I learned frontend development, database integration, authentication systems, dashboard management, and real-time application workflows. This internship improved my technical skills, problem-solving abilities, and understanding of industry-level project development.

I sincerely thank UpSkill Campus, UniConverge Technologies Pvt Ltd (UCT), mentors, and everyone who guided and supported me during this internship journey.



I would like to advise my juniors and peers to focus on practical learning, stay consistent, and continuously improve their technical and problem-solving skills. Working on real-world projects and internships helps in gaining confidence, industry exposure, and valuable experience for future career growth.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



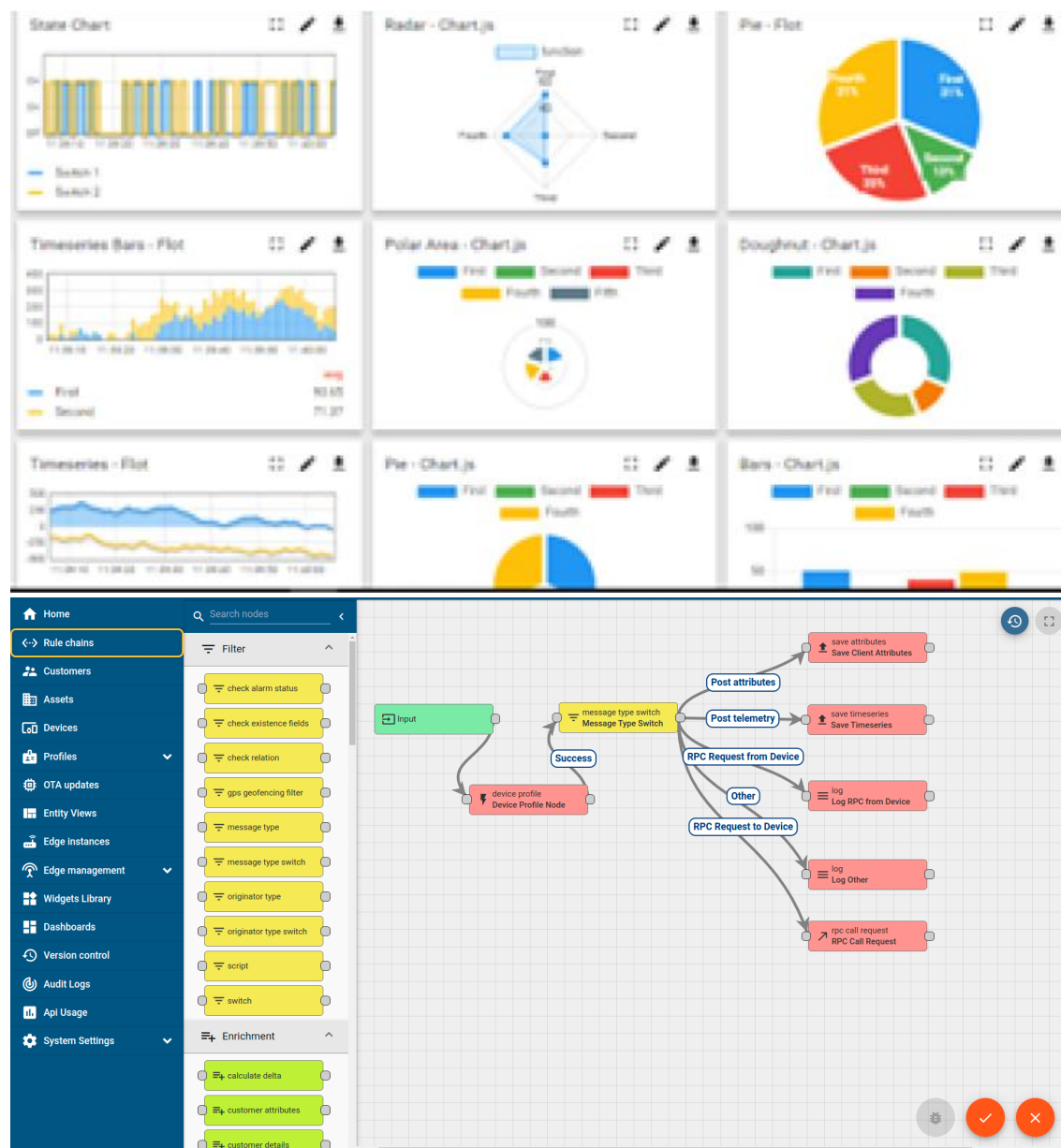
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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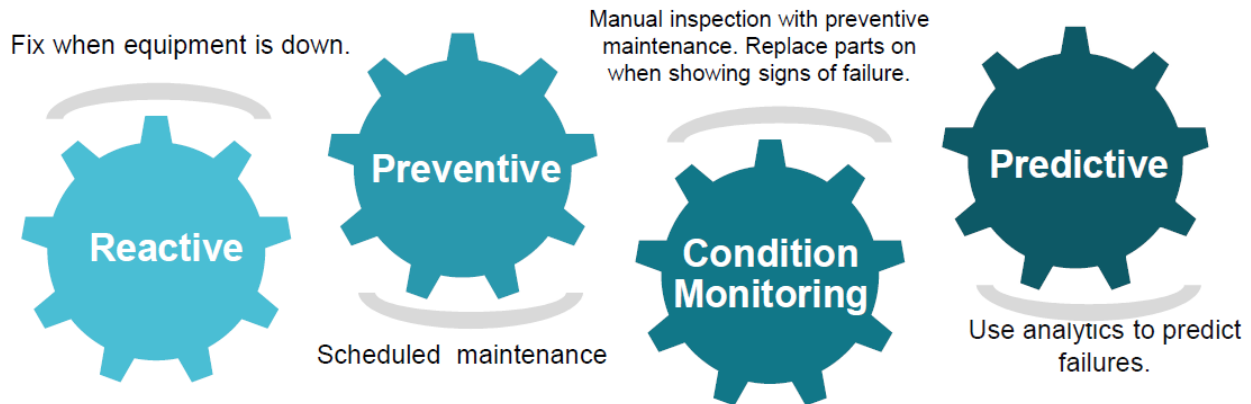


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

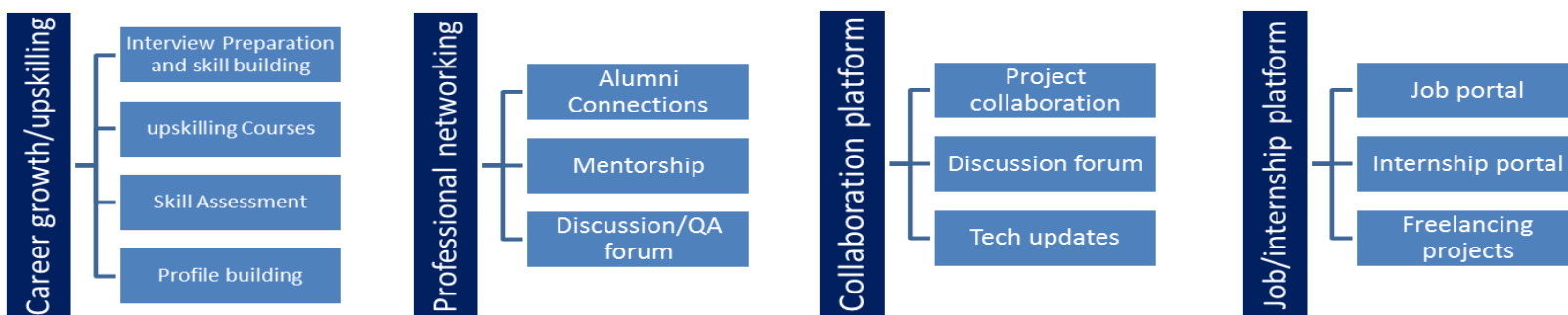
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ▣ get practical experience of working in the industry.
- ▣ to solve real world problems.
- ▣ to have improved job prospects.
- ▣ to have Improved understanding of our field and its applications.
- ▣ to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] React Official Documentation – <https://react.dev/>
- [2] Supabase Documentation – <https://supabase.com/docs>
- [3] Tailwind CSS Documentation – <https://tailwindcss.com/docs>
- [4] Vite Documentation – <https://vitejs.dev/guide/>
- [5] TypeScript Documentation – <https://www.typescriptlang.org/docs/>

2.6 Glossary

Terms	Acronym	Terms	Acronym
Internet of Things	IoT	Application Programming Interface	API
User Interface	UI	Content Delivery Network	CDN
User Experience	UX	Authentication	Auth
Real-Time Database	RTDB	Global Positioning System	GPS
Structured Query Language	SQL	PostgreSQL Database	PostgreSQL

3 Problem Statement

The rapid growth of online food ordering services has increased the need for efficient digital platforms that can connect customers, restaurants, delivery partners, and administrators through a single system. Traditional food ordering methods often face problems such as delayed communication, inefficient order management, lack of real-time tracking, and poor coordination between restaurants and delivery personnel.

The objective of this project is to develop a modern Food Delivery Application that provides a smooth and user-friendly experience for all users. The system allows customers to browse restaurants, place orders, track deliveries in real-time, manage favourites, and submit reviews. Restaurant owners can manage menus and process orders, while delivery partners can handle delivery requests and update delivery status through dedicated dashboards.

The application also focuses on realtime synchronization, role-based authentication, responsive user interfaces, and secure database management using modern web technologies. The proposed system aims to improve efficiency, reduce manual effort, and provide a reliable and scalable platform for food ordering and delivery management.

4 Existing and Proposed solution

Many existing food delivery platforms provide online food ordering and delivery services. However, several systems face limitations such as delayed order updates, lack of realtime delivery tracking, complex user interfaces, limited role-based management, and poor coordination between restaurants and delivery partners. Some applications also have limited customization options and high dependency on manual order handling processes.

The proposed solution is a modern Food Delivery Application developed using React, TypeScript, Tailwind CSS, and Supabase PostgreSQL. The system connects customers, restaurant owners, delivery partners, and administrators through a single platform.

Customers can browse restaurants, place orders, track deliveries in realtime, manage favourites, and submit reviews. Restaurant owners can manage menus, process orders, and monitor restaurant activities through a dedicated dashboard. Delivery partners can accept delivery requests and update delivery status, while administrators can manage users, restaurants, and platform analytics.

The project focuses on realtime synchronization, responsive user interfaces, secure authentication, and role-based access management. Additional value additions include realtime notifications, dynamic delivery tracking, dashboard-based management systems, review and rating functionality, favourites management, and scalable cloud-based architecture.

Future enhancements may include AI-based recommendations, IoT-enabled delivery tracking, smart notifications, and advanced analytics for improving operational efficiency and user experience.

4.1 Code submission Link:

4.2 Report submission Link :

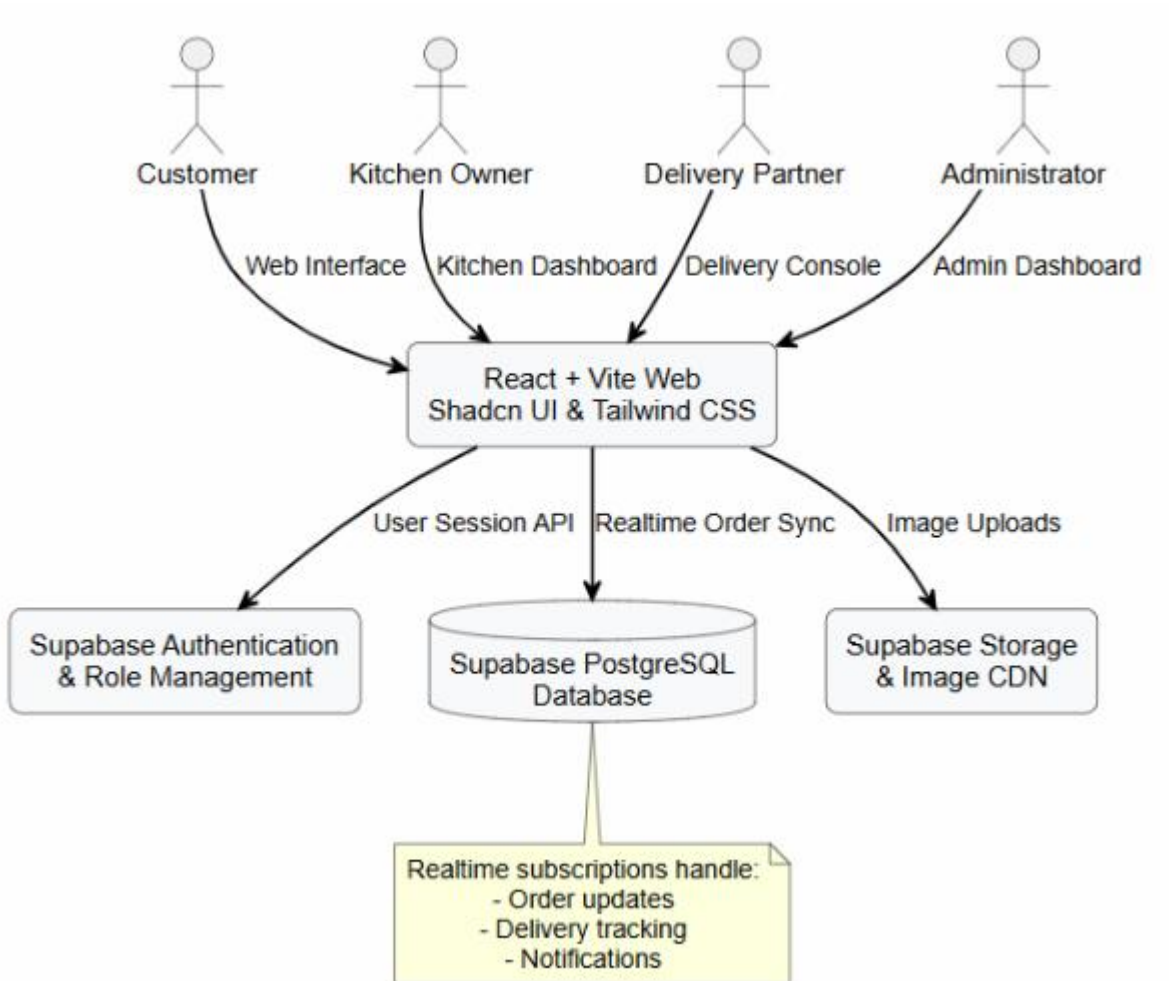
5 Proposed Design/ Model

The proposed Food Delivery Application is designed using a modern full stack web architecture that connects customers, restaurant owners, delivery partners, and administrators through a centralized platform. The system is developed using React, TypeScript, Tailwind CSS, and Supabase PostgreSQL for realtime database management and authentication.

The application follows a role-based workflow where customers can browse restaurants, place orders, track deliveries, and submit reviews. Restaurant owners can manage menus and process orders through a dedicated dashboard, while delivery partners handle delivery requests and update delivery status. Administrators monitor users, restaurants, and platform activities.

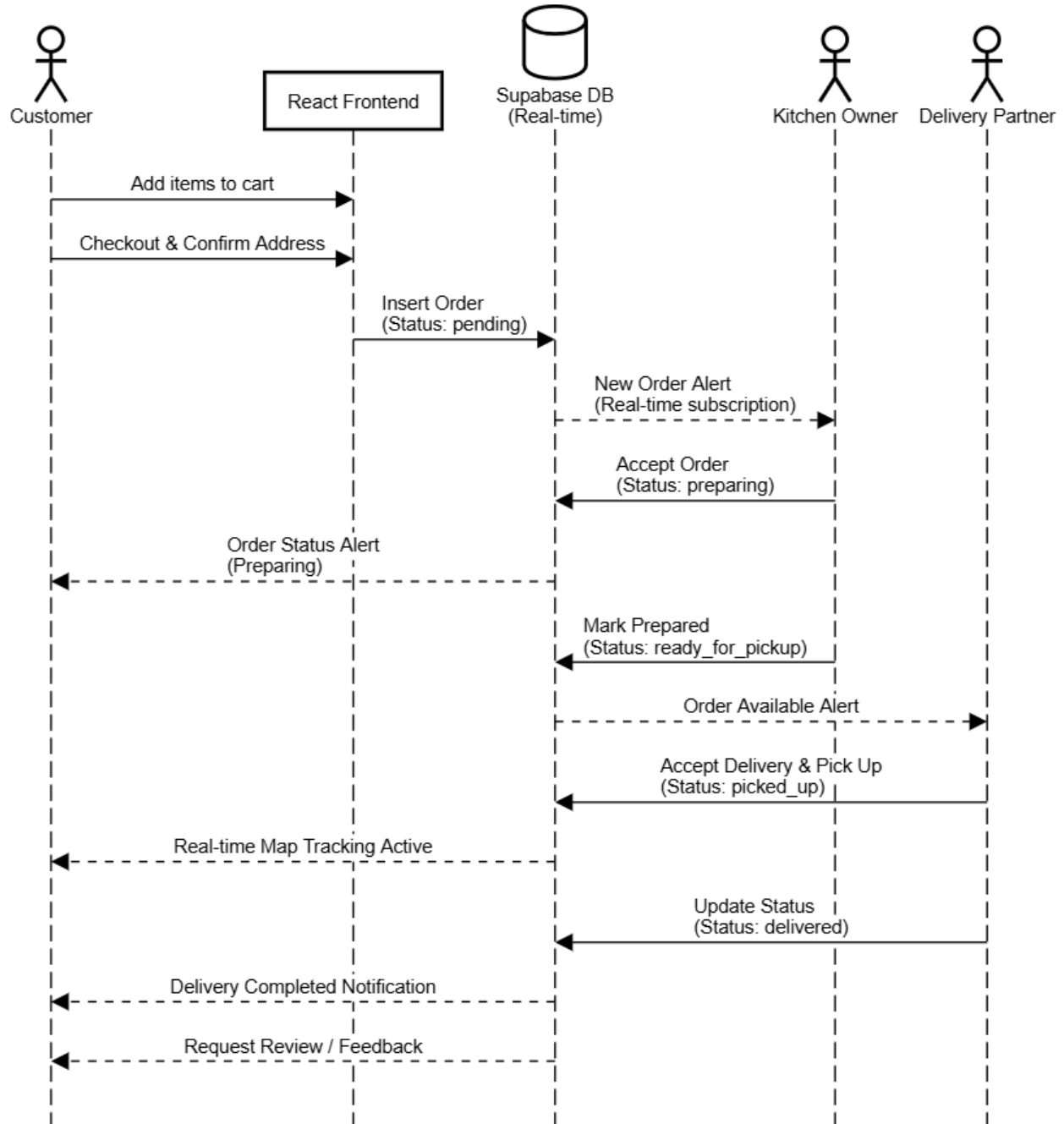
The system design focuses on realtime synchronization, responsive user interfaces, secure authentication, cloud-based data management, and efficient communication between all modules. The following diagrams represent the high-level architecture, low-level workflow, and interface interactions of the proposed system.

5.1 High Level Diagram

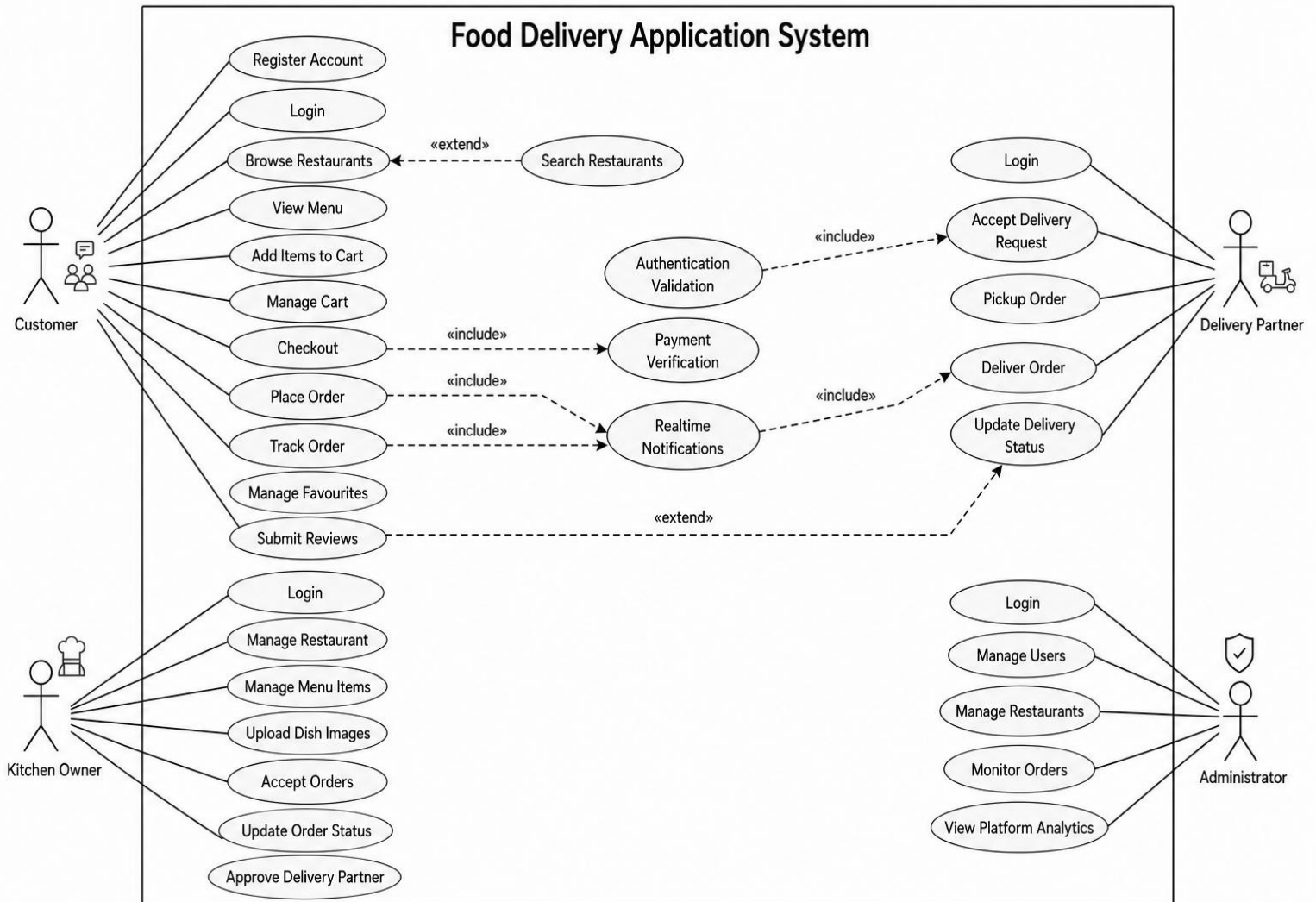


5.2 Low Level Diagram

Food Delivery Workflow Sequence Diagram



5.3 Interfaces (Use Case Diagram)



6 Performance Test

The Food Delivery Application was tested to ensure smooth performance, realtime synchronization, responsive user interaction, and secure role-based access management. The major constraints considered during development included realtime order updates, database synchronization, multiple user-role handling, responsive UI rendering, and secure authentication.

Constraints such as memory usage, realtime database communication, system responsiveness, scalability, and user data security were carefully considered during the system design process. To handle these constraints, Supabase realtime services and PostgreSQL database management were used for efficient cloud-based synchronization and secure data handling. React and Tailwind CSS were used to create responsive and optimized user interfaces for different devices and screen sizes.

Performance testing was conducted by verifying user authentication, restaurant browsing, cart management, realtime order tracking, dashboard operations, and delivery status updates. The application was also tested for smooth navigation, proper role-based access control, and responsive UI behavior under different workflows.

The test results showed that the system successfully handled realtime order synchronization, secure authentication, responsive rendering, and efficient database operations with minimal delays. The application demonstrated stable performance and provided reliable user experience suitable for modern food delivery management systems.

Although advanced industrial-scale load testing and hardware-level performance testing were not performed, the identified constraints and their impact on the system design were analyzed carefully. Future improvements may include advanced scalability optimization, cloud deployment enhancements, and IoT-based realtime tracking features for better operational efficiency.

6.1 Test Plan/ Test Cases

The testing process was planned to verify the functionality, reliability, responsiveness, and security of the Food Delivery Application. Different modules of the system such as authentication, restaurant browsing, cart management, order processing, realtime tracking, dashboard management, and delivery workflows were tested individually and collectively.

The major test cases included:

- User Registration and Login
- Restaurant Search and Menu Browsing
- Add to Cart and Checkout

- Order Placement and Tracking
- Realtime Order Status Updates
- Delivery Partner Workflow
- Restaurant Dashboard Operations
- Administrator Dashboard Access
- Reviews and Favourites Management
- Responsive UI Testing on Multiple Devices

6.2 Test Procedure

The application was tested by executing various user workflows under different user roles including Customer, Kitchen Owner, Delivery Partner, and Administrator. Functional testing was performed to verify proper navigation, authentication, order processing, and real time synchronization.

Responsive testing was conducted on desktop and mobile screen sizes to ensure proper UI rendering. Database operations and realtime updates were tested using Supabase PostgreSQL services. The system was also verified for role-based access control, secure authentication, and dashboard functionality.

6.3 Performance Outcome

The performance testing results showed that the Food Delivery Application provided stable and responsive behavior during user interactions and realtime operations. Authentication, order placement, dashboard updates, and realtime notifications worked successfully with minimal delays.

The system handled realtime database synchronization efficiently and maintained smooth navigation across different modules. Responsive UI performance and secure role-based access control were also successfully verified during testing. Overall, the application demonstrated reliable performance suitable for modern food delivery management systems.

7 My learnings

During this internship, I gained practical knowledge in full stack web development, realtime database management, authentication systems, responsive UI design, and dashboard-based application development. I learned how to design and develop a modern Food Delivery Application using React, TypeScript, Tailwind CSS, and Supabase PostgreSQL.

The internship improved my understanding of frontend development, role-based access control, realtime synchronization, cloud database integration, and application workflow management. I also developed problem-solving skills, debugging techniques, project planning abilities, and experience in working on industry-oriented projects.

This internship experience helped me improve both my technical and professional skills, which will support my future career growth in software development and full stack application engineering.

8 Future work scope

The current Food Delivery Application can be further improved by adding advanced features and industrial-level enhancements in future versions. Due to time limitations, some functionalities could not be fully implemented during the internship period.

Future improvements may include secure online payment gateway integration using services such as Razorpay or Stripe for supporting UPI, debit cards, credit cards, and digital wallet transactions. The system can also be enhanced with realtime delivery tracking using GPS-based location services and mapping APIs such as Google Maps API. This would allow customers to monitor the live location of delivery partners and receive dynamic estimated delivery times.

Additional future enhancements may include AI-based food recommendation systems, IoT-enabled delivery monitoring, smart notification systems, chatbot support, advanced analytics dashboards, cloud scalability optimization, and mobile application deployment for Android and iOS platforms.

The project can also be extended with advanced security mechanisms, automated testing pipelines, and microservice-based architecture to support large-scale commercial deployment in the future.